



Kaushik Muduchuru

Climate Studies PhD

- ▶ IIT Bombay
- ▶ Weather Climate modeller
- ▶ married

Skills

Python (Data Science) 6+ yrs.

**WRF-Chem (RCM),
ECHAM-HAM (GCM)** 4+ yrs.

Shell scripting 4+ yrs.

Scientific writing 3+ yrs.

Education

Jul/2017 - Present

Climate Studies (Ph.D.)

Indian Institute of Technology/
Bombay

weather forecasting ■ *cloud micro-physics*

Jul/2013 - Aug/2015

Earth Sciences (M. Tech.)

Indian Institute of Science/
Bengaluru

Biography

I am a climate/weather researcher passionate about working on real world solutions to complex techno-social problems. I am a chemical engineer by training with graduate work primarily in atmospheric sciences with familiarity in geophysics, geology and signal processing. My work experience from under-graduate days have revolved on environmental problems owing to my childhood days in a town plagued by chrome (from leather tanning) related pollution.

Work experience

Senior Research Fellow | PhD student

Jul/2017 - today

Climate Studies (Prof. Chandra Venkataraman) Indian Institute of Technology - Bombay Dr Vikas Singh and Dr. Amit Kesarkar National Atmospheric Research Laboratory

- Aerosol impact on cloud systems and ensuing rainfall formation processes.
- Climate and weather models where the assimilated meteorological forecasts/output are coupled with aerosols/chemistry.

Skills: Numerical Weather Prediction, Time Series Analysis, Shell scripting

Project Research Fellow

2016 - 2017

Prof. Chandra Venkataraman, Indian Institute of Technology

Disentangling the role of ocean-atmosphere interactions and aerosol fast adjustments on the atmosphere.

Skills: Data analysis and climate modelling

Junior Research Fellow

2015 - 2016

Centre for Atmospheric and Oceanic Sciences (Prof. Jai Sukhatme) Indian Institute of Science - Bengaluru

High resolution outgoing longwave radiation (OLR) data over the Indian region is analyzed with two main objectives:

- Characterize space-time scaling properties of the deep convective systems leading to heavy floods.
- Characterization of northward propagating boreal summer intra-seasonal oscillation (BSISO)

A preliminary conjecture is that Taylor's frozen field hypothesis may hold at both scales (Sub Diurnal and Intraseasonal) and the $-5/3$ scaling in both, frequency and wavenumber domains is a result of the presence of a large scale mean flow.

Skills: Spectral analysis of spatio-temporal data, Signal processing

Teaching Assistant

2017 - Present

Department of chemical engineering, Climate Studies, Indian Institute of Technology

- CL254 (2020) - Process Fluid Mechanics
- CL710 (2020) - Aerosol Technology
- CM802 (2021) - Atmosphere and Climate Change

spectral analysis ■ climate dynamics

Thesis: „Northward propagation of Intra-Seasonal Oscillations: study using high-resolution geostationary satellite data“.

Jun/2009 - May/2013

Chemical Engineering (B.Tech.)

National Institute of Technology - Tiruchirappalli

CFD ■ transport phenomenon

Bachelor's thesis: „Styrene production“.

Languages

Telugu L1

English C1

Tamil, Hindi L2

Contact

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✉ kaushik.reddy.m@gmail.com

🏠 linkedin.com/kaushik-muduchuru

🌐 github.com/Kaushikreddym

Interests

- ▶ Cycling (Downhill Hardtail)
- ▶ Cooking - Mughlai Chettinad

Awards and Affiliations

- Student member of American Geophysical Union
- Graduate Aptitude Test in Engineering [GATE] Postgraduate fellowship
- CLRI-CSIR Summer Fellow

Coursework and related skills

- Graduate Courses: Signal Processing, Statistics, Geophysical Fluid Dynamics, Aerosol Technology, Atmospheric Dynamics, Numerical Methods, Geodynamics, Mantle Convection, Basic Geology, Lithosphere Dynamics, Geochemistry
- Undergraduate Courses: Mass Transfer, Fluid Mechanics, Design of process equipments, dynamical systems in chemical engineering.

Publications

- Muduchuru, K., Venkataraman, C., et. al., (2021) Sensitivity of aerosol feedback effects and chemical mechanisms on the regional scale prediction of cloud properties. in-preparation
- Bhattacharya, A., Muduchuru, K., Venkataraman, C., and Mondal, A., (2021) Anthropogenic aerosols induce drying trends in Indian monsoon wet and dry extremes. submitted to *Nature Climate Change*
- Muduchuru, K. and Venkataraman C (2021). Spatial heterogeneity of aerosol induced rapid adjustments on precipitation response over India : A general circulation model study with ECHAM6-HAM2. *Climate Dynamics*, Jul 30:1-2, pp.1-12.
- Bhattacharya, A., Venkataraman, C., Maity, S. and Muduchuru, K., (2020), December. Analysis of Aerosol Life Cycles over Indian Region in the ECHAM6-HAM2. In *AGU Fall Meeting Abstracts* (Vol. 2020, pp. A063-0003).
- Muduchuru, K. and Venkataraman, C., 2020, December. Spatial heterogeneity of stratiform precipitation response to aerosol fast adjustments over India: A general circulation model study with ECHAM6-HAM2. In *AGU Fall Meeting Abstracts* (Vol. 2020, pp. A063-0016).
- Venkataraman, C., Sharma, A., Tibrewal, K., Maity, S. and Muduchuru, K. (2020), Carbonaceous Aerosol Emissions Sources. *The Magazine for Environmental Managers*
- Patil, N., Venkataraman, C., Muduchuru, K., Ghosh, S. and Mondal, A., (2019). Disentangling sea-surface temperature and anthropogenic aerosol influences on recent trends in South Asian monsoon rainfall. *Climate Dynamics*, 52(3), pp.2287-2302.

Bombay, 7th November 2021

Kaushik Muduchuru